

2-2 Limit of a Function and Limit Laws

師大工教一

※Intuitive Definition of a Limit

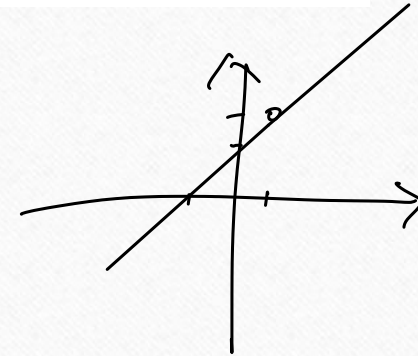
We write $\lim_{x \rightarrow a} f(x) = L$ and say

“the limit of $f(x)$, as x approaches a , equals L ”

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by restricting x to be sufficiently close to a (on either side of a) but not equal to a .

Ex1(p76): How does the function $f(x) = \frac{x^2 - 1}{x - 1}$ behave near $x = 1$?

$$(1) x \neq 1 \quad f(x) = x + 1$$



$$(2) x = 1 \quad f(1) \text{ DNE (does not exist)}$$

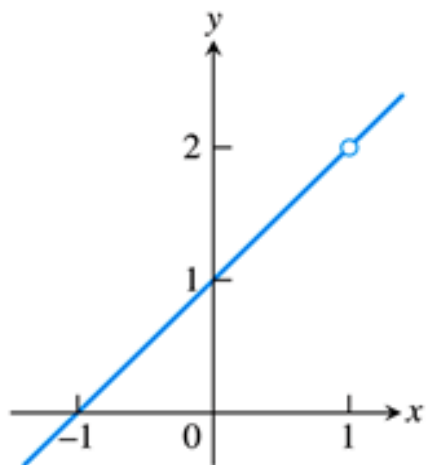
Q: $\lim_{x \rightarrow 1} f(x) = 2$

極限值和求的值無關

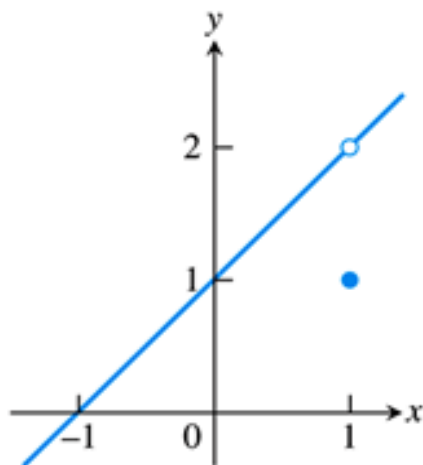
Ex2(p77)

$\lim_{x \rightarrow 1} g(x) = 2$

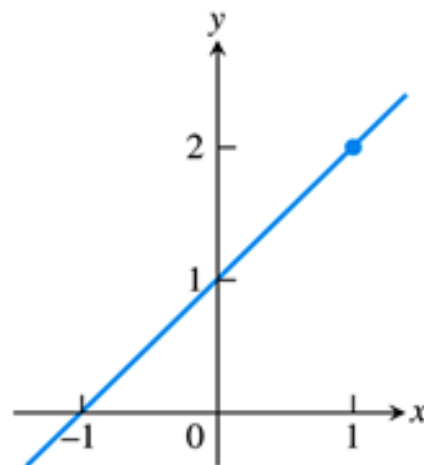
$\lim_{x \rightarrow 1} h(x) = 2$



(a) $f(x) = \frac{x^2 - 1}{x - 1}$

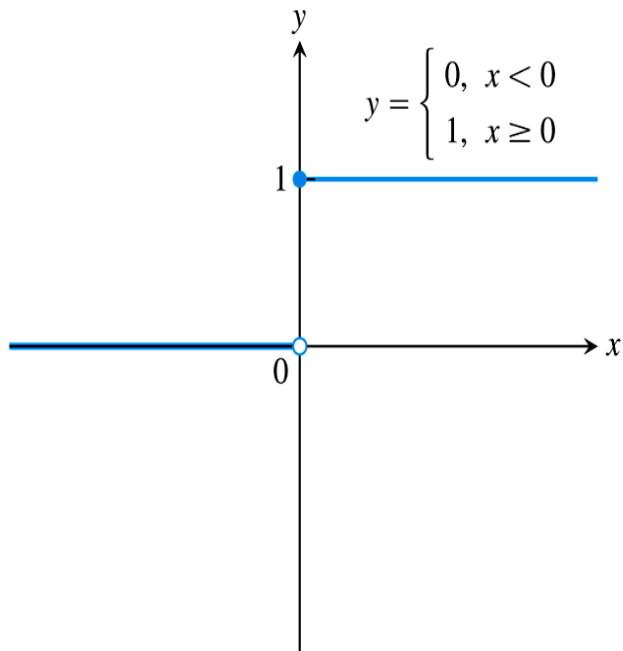
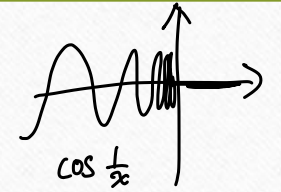


(b) $g(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$

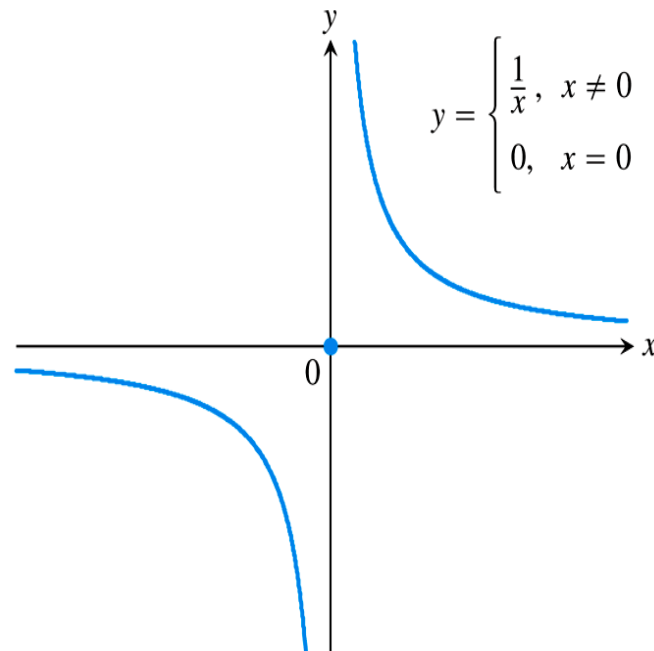


(c) $h(x) = x + 1$

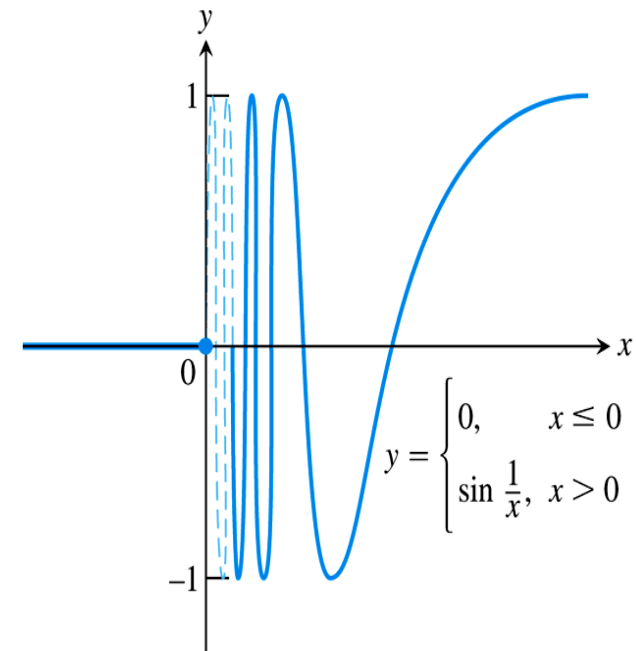
Ex4(p78): The types that the limits do not exist.



(a) Unit step function $U(x)$



(b) $g(x)$



(c) $f(x)$

FIGURE 2.10 None of these functions has a limit as x approaches 0 (Example 4).

※The Limit Law

Suppose that c is a constant and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists.

常數

$$(1) \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) \quad (2) \lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$$

$$(3) \lim_{x \rightarrow a} [f(x) g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) \quad (4) \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

$$(5) \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \quad (6) \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$$

$$(7) \lim_{x \rightarrow a} c = c, \lim_{x \rightarrow a} x = a \quad (8) \lim_{x \rightarrow a} x^n = a^n \quad (9) \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$$

Ex5(p79): Find (c) $\lim_{x \rightarrow -2} \sqrt{4x^2 + 3}$

$$\lim_{x \rightarrow -2} \sqrt{4x^2 + 3} = \sqrt{\lim_{x \rightarrow -2} (4x^2 + 3)}$$

$$= \sqrt{\left(\lim_{x \rightarrow -2} 4x^2\right) + \left(\lim_{x \rightarrow -2} 3\right)}$$

$$= \sqrt{4\left(\lim_{x \rightarrow -2} x\right)^2 + 3}$$

$$= \sqrt{4(-2)^2 + 3} = 17$$

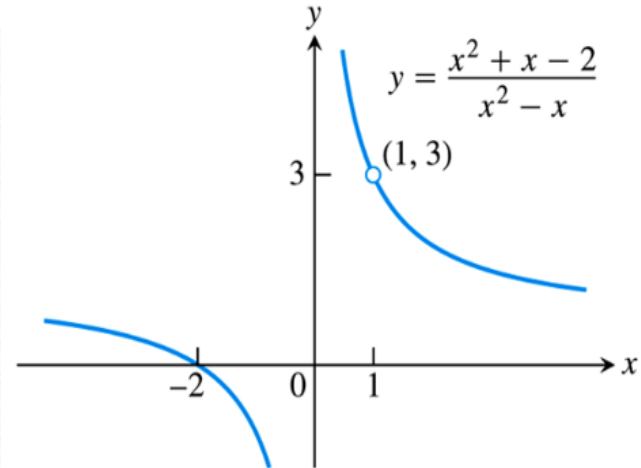
rational function 有理函数 $R(x) = \frac{p(x)}{q(x)}$ where p, q are polynomials.

Theorem: If f is a polynomial, a rational function, an n th root of a polynomial, or an n th root of a rational function, and a is in the domain of f , then $\lim_{x \rightarrow a} f(x) = f(a)$.

Ex7(p81): Evaluate $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}$

$$\text{法①} = \lim_{x \rightarrow 1} \frac{2x+1}{2x-1} = \frac{3}{1} = 3$$

$$\text{法②} = \lim_{x \rightarrow 1} \frac{(x+2)\cancel{(x-1)}}{x\cancel{(x-1)}} = \frac{3}{1} = 3$$



※The Sandwich Theorem 夾擠定理 (Squeeze Theorem)

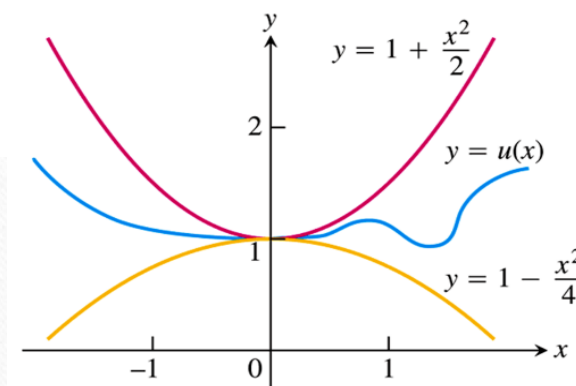
If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L \text{ then } \lim_{x \rightarrow a} g(x) = L$$

Ex10(p83): Given a function u that satisfies $1 - \frac{x^2}{4} \leq u(x) \leq 1 + \frac{x^2}{2}$ for all

$x \neq 0$, find $\lim_{x \rightarrow 0} u(x)$.

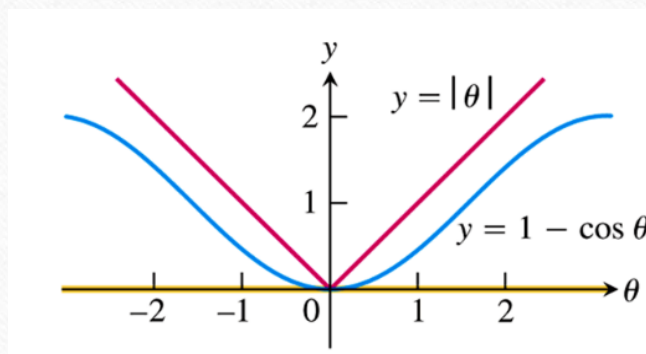
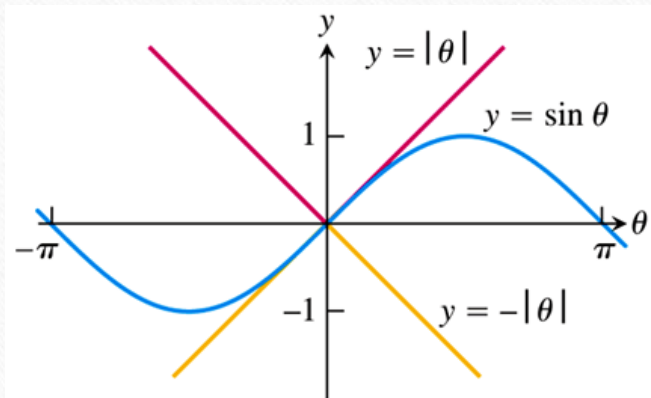
$$\begin{aligned} \lim_{x \rightarrow 0} 1 - \frac{x^2}{4} &= 1 \\ \lim_{x \rightarrow 0} 1 + \frac{x^2}{2} &= 1 \\ \text{By Sandwich Theorem} \\ \lim_{x \rightarrow 0} u(x) &= 1 \end{aligned}$$



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Ex11(p83): Show that (a) $\lim_{\theta \rightarrow 0} \sin \theta = 0$ (b) $\lim_{\theta \rightarrow 0} \cos \theta = 1$

(c) For any function f , $\lim_{x \rightarrow c} |f(x)| = 0$ implies $\lim_{x \rightarrow c} f(x) = 0$.



(a) $\sin \theta \rightarrow -|\theta| \leq \sin \theta \leq |\theta|$
 $\lim_{\theta \rightarrow 0} \sin \theta = 0$
 $\lim_{\theta \rightarrow 0} -|\theta| = 0$, $\lim_{\theta \rightarrow 0} |\theta| = 0$
 By Sandwich Theorem
 $\lim_{\theta \rightarrow 0} \sin \theta = 0$

(b) $\cos \theta \rightarrow 0 \leq 1 - \cos \theta \leq |\theta|$
 $\lim_{\theta \rightarrow 0} 0 = 0$, $\lim_{\theta \rightarrow 0} |\theta| = 0$
 $\Rightarrow \lim_{\theta \rightarrow 0} (1 - \cos \theta) = 0$

$\lim_{\theta \rightarrow 0} \cos \theta = \lim_{\theta \rightarrow 0} 1 - (1 - \cos \theta)$
 $= \lim_{\theta \rightarrow 0} 1 - (\lim_{\theta \rightarrow 0} 1 - \lim_{\theta \rightarrow 0} \cos \theta) = 1$

(c) $\lim_{x \rightarrow c} |f(x)| = 0$
 $\lim_{x \rightarrow c} -|f(x)| = 0$
 by Sandwich Theorem
 $\lim_{x \rightarrow c} f(x) = 0$

HW2-2

- HW: 14,18,24,36,47,54,66,80,81

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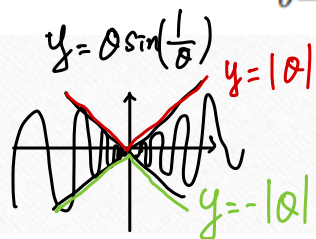
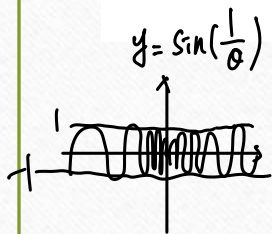
2. (18 分) 計算下列極限值。

$$\frac{((7x)^{\frac{1}{2}} - 7)'}{(7 - x)'} = \frac{(\frac{1}{2}x)^{-\frac{1}{2}}}{-1} = -\frac{1}{\sqrt{\frac{1}{2}x}} = -\frac{\sqrt{2}}{\sqrt{x}}$$

V(a) $\lim_{\theta \rightarrow 0} \theta \sin(\frac{1}{\theta})$

(b) $\lim_{x \rightarrow 7} \frac{\sqrt{7x} - 7}{7 - x}$

(c) $\lim_{t \rightarrow -\infty} \frac{5 - 4t^3}{3t^2 + 2}$



$$-1 \leq \sin \frac{1}{\theta} \leq 1$$

$$-|\theta| \leq \theta \sin \frac{1}{\theta} \leq |\theta|$$

$$\lim_{\theta \rightarrow 0} -|\theta| = 0 = \lim_{\theta \rightarrow 0} |\theta|$$

By Sandwich Theorem
 $\lim_{\theta \rightarrow 0} \theta \sin(\frac{1}{\theta}) = 0$

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1. (25 pts) Find the limit (極限).

(a) $\lim_{\theta \rightarrow 0^+} \operatorname{arccot}(\frac{1}{\theta}) \sin^2(\frac{1}{\theta})$

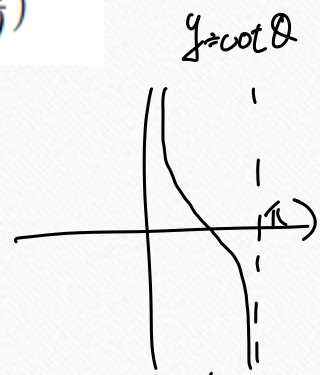
$$0 \leq \sin^2(\frac{1}{\theta}) \leq 1$$

$$0 \leq \operatorname{arccot}(\frac{1}{\theta}) \sin^2(\frac{1}{\theta}) \leq |\operatorname{arccot}(\frac{1}{\theta})|$$

$$\lim_{\theta \rightarrow 0^+} 0 = 0 = \lim_{\theta \rightarrow 0^+} |\operatorname{arccot}(\frac{1}{\theta})|$$

By Sandwich Theorem

$$\lim_{\theta \rightarrow 0^+} \operatorname{arccot}(\frac{1}{\theta}) \sin^2(\frac{1}{\theta}) = 0$$



$$\theta \rightarrow 0^+ \Rightarrow \cot \theta \rightarrow \infty$$

$$\frac{1}{\theta} \rightarrow \infty \Rightarrow \operatorname{arccot}(\frac{1}{\theta}) \rightarrow 0$$

2-3 The Precise Definition of a Limit

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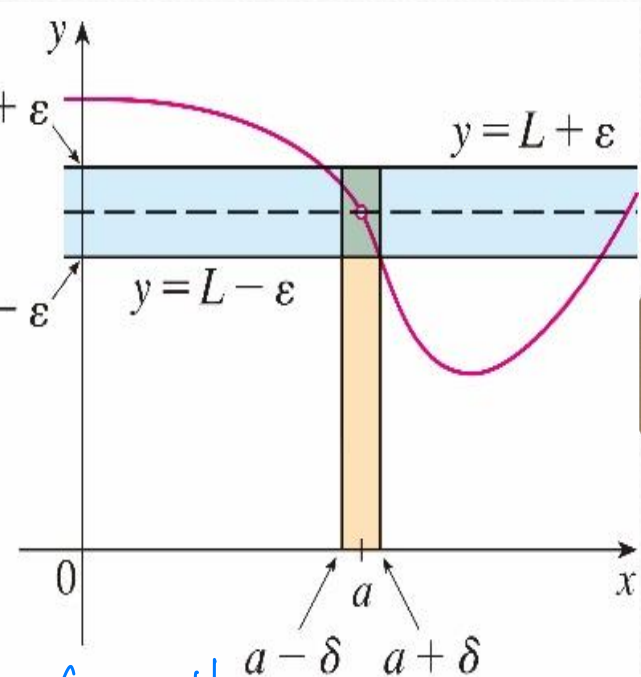
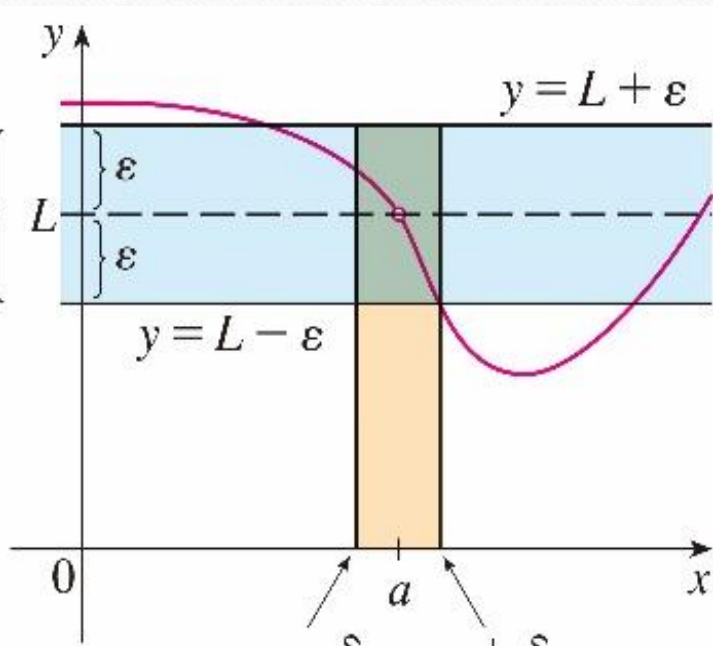
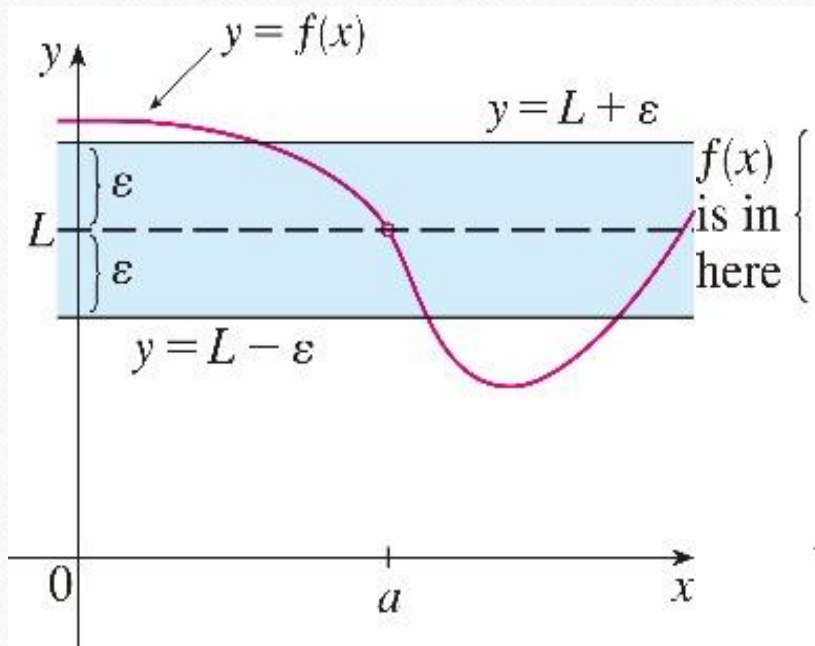
2-3 The Precise Definition of a Limit

DEFINITION Let $f(x)$ be defined on an open interval about c , except possibly at c itself. We say that the **limit of $f(x)$ as x approaches c is the number L** , and write

$$\lim_{x \rightarrow c} f(x) = L,$$

if, for every number $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - c| < \delta.$$



$\forall \varepsilon > 0, \exists \delta > 0$ such that
as $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$

$\forall$: for all
 $\exists$: there exists
存在

Ex2(p89) Show that $\lim_{x \rightarrow 1} (5x - 3) = 2$.

analysis

$$|(5x-3)-2| < \epsilon$$

$$\Leftrightarrow |5x-5| < \epsilon$$

$$\Leftrightarrow 5|x-1| < \epsilon$$

$$\Leftrightarrow |x-1| < \frac{\epsilon}{5}$$

$$\forall \epsilon > 0$$

$$\text{Let } \delta = \frac{\epsilon}{5}$$

$$\text{As } 0 < |x-1| < \delta = \frac{\epsilon}{5}$$

$$5|x-1| < \epsilon$$

$$|5x-5| < \epsilon$$

$$|(5x-3)-2| < \epsilon$$

$$\therefore \lim_{x \rightarrow 1} (5x-3) = 2$$

HW2-3

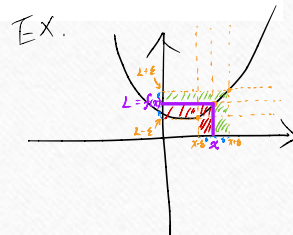
-
- No HW.

110 9. (加分題)

(a) State the definition of limit $\lim_{x \rightarrow x_0} f(x) = l$. (2 pts)

(b) Prove that $\lim_{x \rightarrow 2} (3x + 1) = 7$ by the definition of limit. (3 pts)

(a) $\forall \varepsilon > 0, \exists \delta > 0$ such that
as $0 < |x - x_0| < \delta \Rightarrow |f(x) - l| < \varepsilon$.



$$(b) \lim_{x \rightarrow 2} (3x + 1) = 7$$

$$|(3x + 1) - 7| < \varepsilon$$

$$\Leftrightarrow |3x - 6| < \varepsilon$$

$$\Leftrightarrow 3|x - 2| < \varepsilon$$

$$\Leftrightarrow |x - 2| < \frac{\varepsilon}{3}$$

$$\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{3} > 0$$

$$\text{such that as } 0 < |x - 2| < \delta = \frac{\varepsilon}{3}$$

$$\Rightarrow 3|x - 2| < \varepsilon$$

$$\Rightarrow |3x - 6| < \varepsilon$$

$$\Rightarrow |(3x + 1) - 7| < \varepsilon$$

$$\therefore \lim_{x \rightarrow 2} (3x + 1) = 7$$

2-4 One-Sided Limits

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2-4 One-Sided Limits

(1) We write $\lim_{x \rightarrow a^+} f(x) = L$ and say the **right-hand limit** of $f(x)$, as x

approaches a , is equal to L (or the limit of $f(x)$, as x approaches a

from the right) is equal to L if we can make the values of $f(x)$ arbitrarily

任意地

close to L (as close to L as we like) by restricting x to be sufficiently close to a **with x greater than a** .

Ex1 (p97)

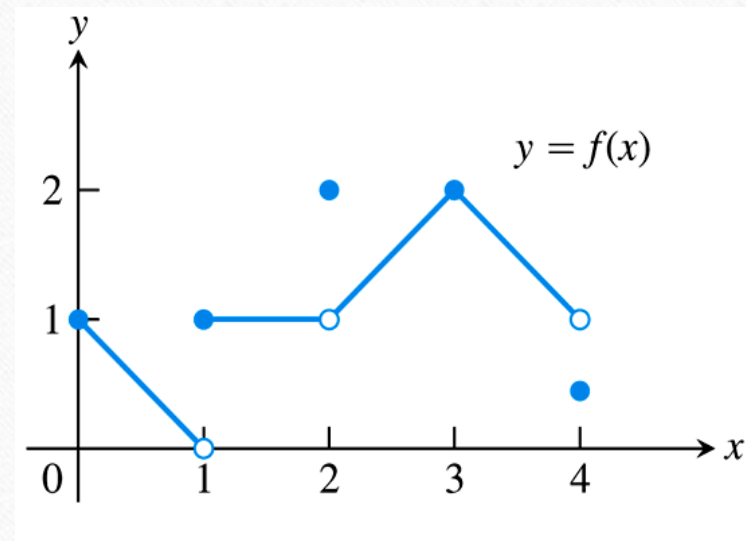
(1) $\lim_{x \rightarrow 0^+} f(x) = 1 = \lim_{x \rightarrow 0} f(x)$, $\lim_{x \rightarrow 0^-} f(x)$ DNE

(2) $\lim_{x \rightarrow 1^+} f(x) = 1$, $\lim_{x \rightarrow 1^-} f(x) = 0$, so $\lim_{x \rightarrow 1} f(x)$ DNE

(3) $\lim_{x \rightarrow 2^+} f(x) = 1$, $\lim_{x \rightarrow 2^-} f(x) = 1$, $\lim_{x \rightarrow 2} f(x) = 1$

(4) $\lim_{x \rightarrow 3^+} f(x) = 2$, $\lim_{x \rightarrow 3^-} f(x) = 2$, $\lim_{x \rightarrow 3} f(x) = 2$

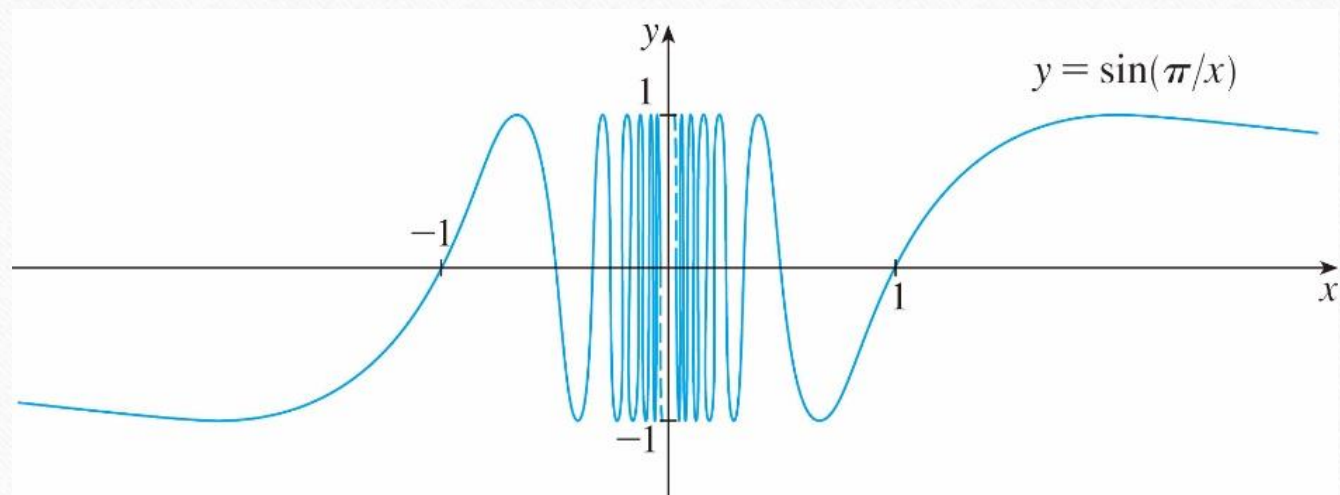
(5) $\lim_{x \rightarrow 4^+} f(x)$ DNE , $\lim_{x \rightarrow 4^-} f(x) = 1 = \lim_{x \rightarrow 4} f(x)$



Ex4(p99): Investigate $\lim_{x \rightarrow 0} \sin \frac{1}{x}$

$$\lim_{x \rightarrow 0^+} \sin \frac{1}{x} \text{ DNE}$$

so is $\lim_{x \rightarrow 0^-} \sin \frac{1}{x}$, $\lim_{x \rightarrow 0} \sin \frac{1}{x}$



Theorem: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ (θ in radians)

Ex5(p100): Show that (1) $\lim_{y \rightarrow 0} \frac{\cos y - 1}{y} = 0$

$$\begin{aligned} (1) &= \lim_{y \rightarrow 0} \frac{(\cos y - 1)(\cos y + 1)}{y(\cos y + 1)} = \lim_{y \rightarrow 0} \frac{\cos^2 y - 1}{y(\cos y + 1)} = \lim_{y \rightarrow 0} \frac{-\sin^2 y}{y(\cos y + 1)} \\ &= \lim_{y \rightarrow 0} \frac{-\sin y}{y} \cdot \frac{\sin y}{\cos y + 1} = \lim_{y \rightarrow 0} \frac{\sin y}{y} \times \lim_{y \rightarrow 0} \frac{\sin y}{\cos y + 1} = -1 \times \frac{0}{1+1} = 0 \end{aligned}$$

(2) $\lim_{x \rightarrow 0} \frac{\sin 2x}{5x} = \frac{2}{5}$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \cdot \frac{2x}{5x} = \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \times \lim_{x \rightarrow 0} \frac{2x}{5x} \\ &= 1 \times \frac{2}{5} = \frac{2}{5} \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{2x} = 1$$

Let $2x = \theta$

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{2x} = \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

HW2-4

- HW: 1,5,6,20,22,25,31,36

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 0$$

$$-1 \leq \sin \theta \leq 1$$

$$\frac{-1}{|\theta|} \leq \frac{\sin \theta}{\theta} \leq \frac{1}{|\theta|}$$

$$\lim_{\theta \rightarrow 0} \frac{-1}{|\theta|} = 0$$

$$\lim_{\theta \rightarrow 0} \frac{1}{|\theta|} = 0$$

by Sandwich Theorem

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 0$$

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\frac{1}{2} \sin \theta < \frac{1}{2} \theta < \frac{1}{2} \tan \theta$$

$$1 < \frac{\theta}{\sin \theta} < \frac{1}{\cos \theta}$$

$$\cos \theta < \frac{\sin \theta}{\theta} < 1$$

$$\lim_{\theta \rightarrow 0} \cos \theta = 1$$

$$\lim_{\theta \rightarrow 0} 1 = 1$$

By Sandwich Theorem

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$



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2. Find the limit if it exists. Notice that L'Hôpital's Rule is forbidden. (20 pts)

$$(a) \lim_{x \rightarrow 2} \frac{\sqrt{x+7} - 3}{x-2}$$

$$\cancel{v(b)} \lim_{x \rightarrow 0} \frac{\tan(5x)}{\sin(3x)}$$

$$(c) \lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 3})$$

$$(d) \lim_{x \rightarrow \infty} \frac{\sqrt[3]{8x^6 + 2x^3 + 1}}{\sqrt{4x^4 + 3x + 1}}$$

$$\begin{aligned} (b) \lim_{x \rightarrow 0} \frac{\tan(5x)}{\sin(3x)} &= \lim_{x \rightarrow 0} \left(\frac{\sin 5x}{\cos 5x} \times \frac{1}{\sin 3x} \right) = \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \times \frac{3x}{\sin 3x} \times \frac{5x}{3x} \times \frac{1}{\cos 5x} \\ &= \lim_{x \rightarrow 0} \left(\frac{\sin 5x}{5x} \times \frac{1}{\frac{\sin 3x}{3x}} \times \frac{5}{3} \times \frac{1}{\cos 5x} \right) \\ &= 1 \times 1 \times \frac{5}{3} \times 1 = \frac{5}{3} \end{aligned}$$

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4. (28 pts) Find the limit (極限).

$$(c) \lim_{\theta \rightarrow 0} \theta^2 \cot(7\theta) \csc(4\theta)$$

$$= \lim_{\theta \rightarrow 0} \theta^2 \cdot \frac{\cos 7\theta}{\sin 7\theta} \cdot \frac{1}{\sin 4\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{\theta^2}{\sin 7\theta} \cdot \frac{4\theta}{\sin 4\theta} \cdot \frac{1}{28}$$

$$= \lim_{\theta \rightarrow 0} \frac{1}{\frac{\sin 7\theta}{7\theta}} \cdot \frac{1}{\frac{\sin 4\theta}{4\theta}} \cdot \frac{1}{28}$$

$$= 1 \times 1 \times \frac{1}{28} = \frac{1}{28}$$

$$(a) \lim_{x \rightarrow -\infty} (2x + \sqrt{4x^2 + 1})$$

$$v(c) \lim_{\theta \rightarrow 0} \theta^2 \cot(7\theta) \csc(4\theta)$$

$$(b) \lim_{x \rightarrow 0^+} \frac{\sqrt{x+1} - 1}{\sqrt{x^2+1} - \sqrt{x+1}}$$

$$(d) \lim_{t \rightarrow 0} \ln \sqrt{(1+t)^{\frac{1}{3t}}}$$

2-5 Continuity

連續性

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※Continuity at a Point

Definition:

(i) The function f is **continuous at** c if $\lim_{x \rightarrow c} f(x) = f(c)$.

(ii) The function f is **right-continuous at** c (or continuous from the right) if $\lim_{x \rightarrow c^+} f(x) = f(c)$.

(iii) The function f is **left-continuous at** c (or continuous from the left) if $\lim_{x \rightarrow c^-} f(x) = f(c)$.

Continuity Test: $\lim_{x \rightarrow c} f(x) = f(c)$ if and only if

1. $f(c)$ exists,

2. $\lim_{x \rightarrow c} f(x)$ exists,

3. $\lim_{x \rightarrow c} f(x) = f(c)$

Ex1(p104): At which point does the function f appear to be not continuous?

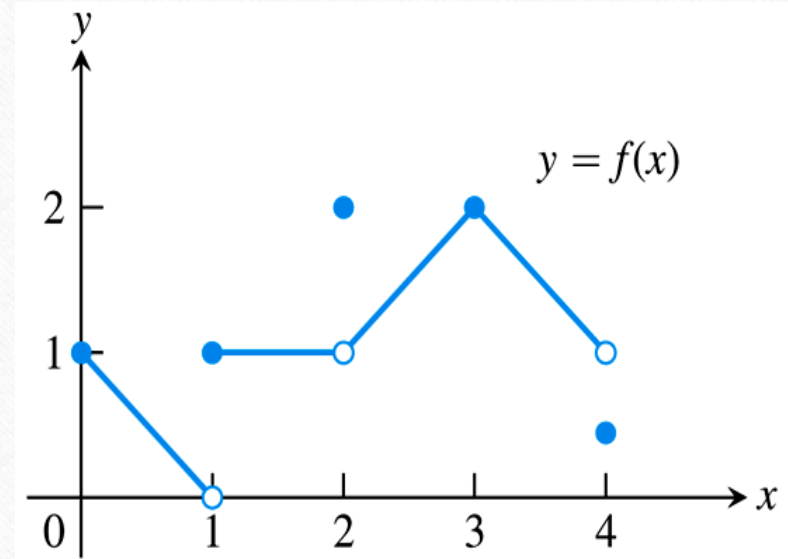
$$\lim_{x \rightarrow 1^-} f(x) = 0 \neq \lim_{x \rightarrow 1^+} f(x) = 1 = f(1)$$

jump discontinuity 跳動型不連續

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = 1 \Rightarrow \lim_{x \rightarrow 2} f(x) = 1$$

$$\text{But } f(2) = 2 \neq \lim_{x \rightarrow 2} f(x)$$

removable discontinuity 可移除不連續



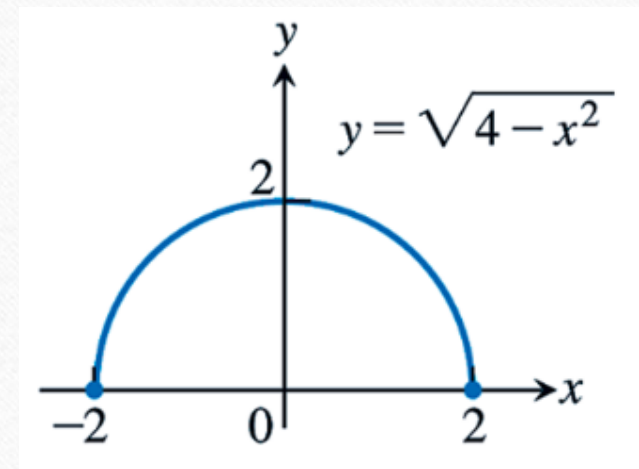
Definition: A function f is continuous over a closed interval $[a, b]$ if it is continuous at all interior points in the interval, right-continuous at a , and left-continuous at b .

Ex2(p105): Show that $f(x) = \sqrt{4-x^2}$ is continuous on $[-2, 2]$.

$$\begin{aligned} (1) \quad a \in (-2, 2) \\ \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} \sqrt{4-x^2} = \sqrt{\lim_{x \rightarrow a} (4-x^2)} \\ &= \sqrt{4-a^2} = f(a) \quad \therefore f \text{ is continuous at } a \end{aligned}$$

$$(2) \quad \lim_{x \rightarrow 2^-} f(x) = 0 = f(2) \quad \therefore f \text{ is continuous at } 2$$

$$(3) \quad \lim_{x \rightarrow -2^+} f(x) = 0 = f(-2) \quad \therefore f \text{ is continuous at } x = -2$$



Ex4(p105): Investigate the continuity of $y = \lfloor x \rfloor$.

$$\lfloor x \rfloor = [x] = \llbracket x \rrbracket = \begin{cases} x, & x \in \mathbb{Z} \\ \text{比 } x \text{ 小的最大整数}, & x \in \mathbb{R} \setminus \mathbb{Z} \end{cases}$$

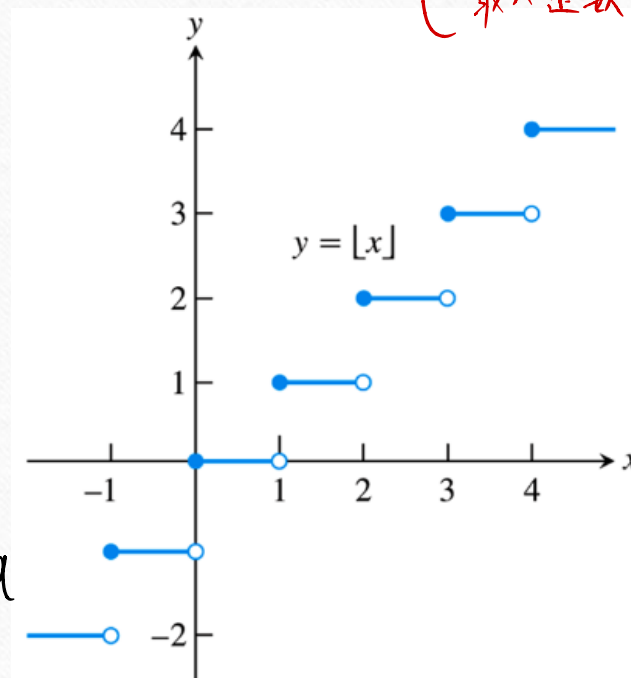
$$n \in \mathbb{Z} \quad \lim_{x \rightarrow n^+} \lfloor x \rfloor = n \neq \lim_{x \rightarrow n^-} \lfloor x \rfloor = n-1$$

$\therefore \lfloor \cdot \rfloor$ is not continuous, at $x=n$

$a \notin \mathbb{Z}$ Assume $n \leq a < n+1$

$$\lim_{x \rightarrow a^+} \lfloor x \rfloor = \lim_{x \rightarrow a^-} \lfloor x \rfloor = n = \lfloor a \rfloor$$

$\therefore \lfloor \cdot \rfloor$ is continuous, at $x=a$

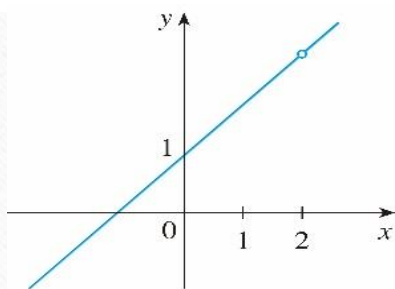


※Types of Discontinous Points

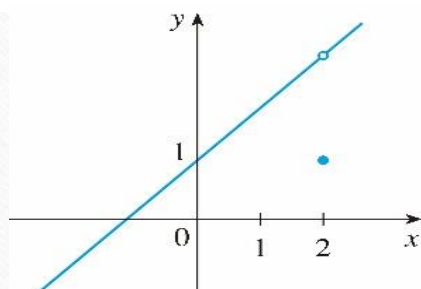
Ex: Where are each of the following functions discontinuous?

$$(a) f(x) = \frac{x^2 - x - 2}{x - 2} \quad (b) f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

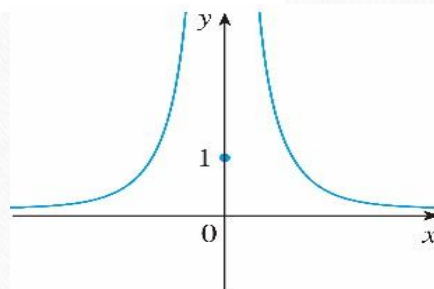
$$(c) f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \quad (d) f(x) = \llbracket x \rrbracket$$



$$(a) f(x) = \frac{x^2 - x - 2}{x - 2}$$

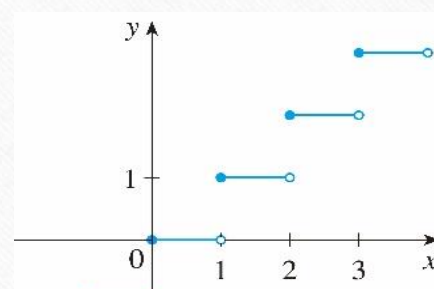


$$(c) f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$



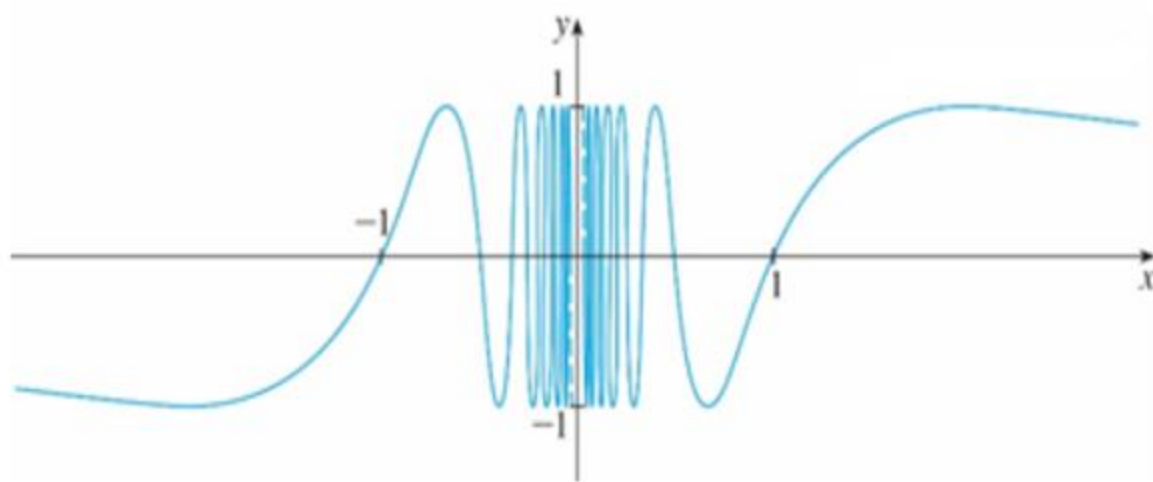
$$(b) f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

無窮型不連續



$$(d) f(x) = \llbracket x \rrbracket$$

Supplement: $f(x) = \sin \frac{1}{x}$



※Propertie of Continuous Functions(p107)

Theorem 8: If f and g are continuous at a and k is a constant, then the following functions are continuous at a

$$1. f \pm g \quad 2. k \cdot f \quad 3. f \cdot g \quad 4. \frac{f}{g} \quad (g(a) \neq 0) \quad 5. f^n \quad 6. \sqrt[n]{f}$$

Theorem: The following types of functions are continuous at every number in their domain:

rational function $R(x) = \frac{P(x)}{Q(x)}$, p, q polynomials

1. polynomials
2. rational functions
3. root functions
4. trigonometric functions
5. inverse trigonometric functions
6. exponents
7. logarithms

Ex 7 (p107) Investigating the continuity of $f(x) = |x|$.

$$f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

(1) $x > 0$ $f(x) = x$ conti.
 (2) $x < 0$ $f(x) = -x$ conti.
 (3) $x = 0$ $\lim_{x \rightarrow 0} |x| = \lim_{x \rightarrow 0} x = 0$

Ex8(p108): The following functions are continuous on their natural domain:

(a) $y = \sqrt{x^2 - 2x - 5}$ (b) $y = \frac{x^{\frac{2}{3}}}{1 + x^4}$ (c) $y = \left| \frac{x-2}{x+2} \right|$ (d) $y = \left| \frac{x \sin x}{x^2 + 2} \right|$

Let $f(x) = \sqrt{x}$, $g(x) = x^2 - 2x - 5$

$\sqrt{x^2 - 2x - 5} = (f \circ g)(x)$

f is a root function. g is a polynomial.

$\therefore f, g$ are conti. $\Rightarrow (f \circ g)(x)$ is conti.

Let $f(x) = |x|$, $g(x) = \frac{x-2}{x+2}$

$\left| \frac{x-2}{x+2} \right| = (f \circ g)(x)$

$\therefore f, g$ are conti. $\Rightarrow f \circ g$ is conti.

Theorem 9—Compositions of Continuous Functions

If f is continuous at c and g is continuous at $f(c)$, then the composition $g \circ f$ is continuous at c .

Ex9(p109) Find

$$(a) \lim_{x \rightarrow \pi/2} \cos \left(2x + \sin \left(\frac{3\pi}{2} + x \right) \right) = \cos \left(\lim_{x \rightarrow \pi/2} \left(2x + \sin \left(\frac{3\pi}{2} + x \right) \right) \right) = \cos \left(\lim_{x \rightarrow \pi/2} 2x + \lim_{x \rightarrow \pi/2} \sin \left(\frac{3\pi}{2} + x \right) \right) = \cos (\pi + 0) = \cos \pi = -1$$

$$(b) \lim_{x \rightarrow 1} \sin^{-1} \left(\frac{1-x}{1-x^2} \right)$$

$$(c) \lim_{x \rightarrow 0} \sqrt{x+1} e^{\tan x}$$

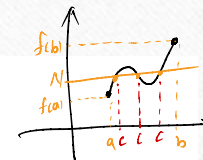
✱ The Intermediate Value Theorem (I.V.T.) 中間值定理

Intermediate Value Theorem

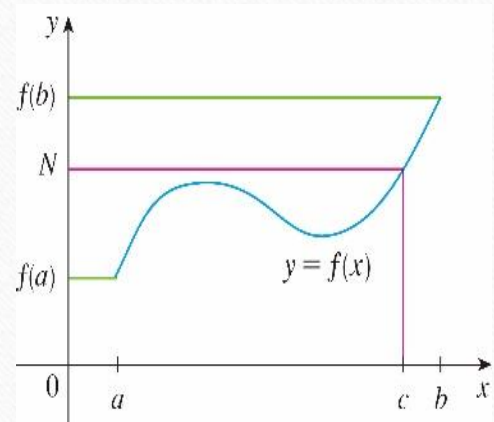
(H1) f is continuous on $[a, b]$.

(H2) N lies between $f(a)$ and $f(b)$.

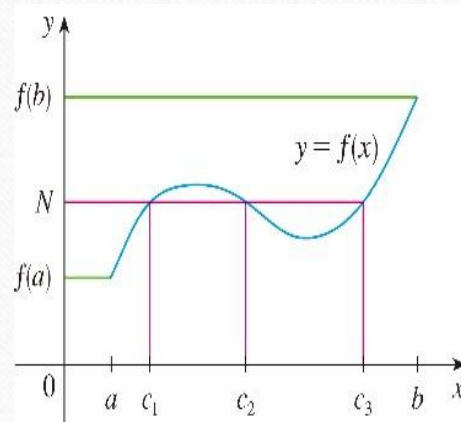
(C) $\exists c \in (a, b)$ such that $f(c) = N$.



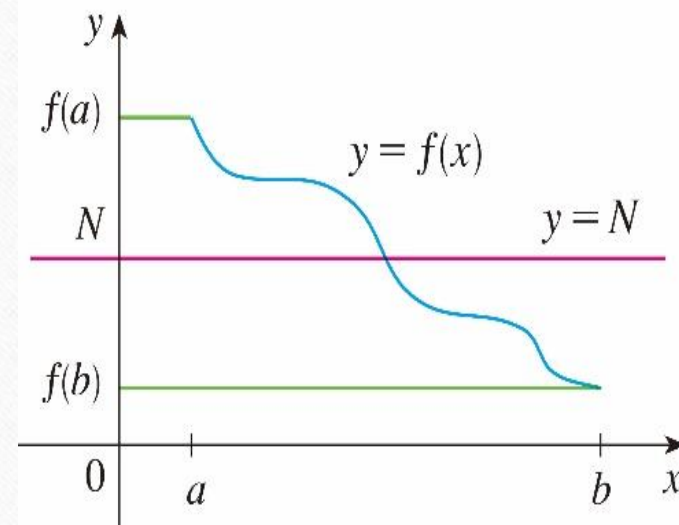
Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.



(a)



(b)



Ex11(p111): Show that the equation $\sqrt{2x+5} = 4 - x^2$ has a solution.

$$\text{Let } f(x) = \sqrt{2x+5} - 4 + x^2$$

$$f(0) = \sqrt{5} - 4 + 0 < 0$$

$$f(2) = \sqrt{9} - 4 + 4 > 0$$

$$\text{Let } g(x) = \sqrt{x}, h(x) = 2x+5, k(x) = 4 - x^2$$

$\therefore g$ is a root function, h is a polynomial

$\therefore \sqrt{2x+5} = (g \circ h)(x)$ is also conti.

$k(x) = -4 + x^2$ is a polynomial

$\therefore f(x) = \sqrt{2x+5} - 4 + x^2$ is also conti. on $[0, 2]$

By I.V.T. $\exists c \in (0, 2)$ such that $f(c) = 0$
 $\sqrt{2c+5} - 4 + c^2 = 0 \Rightarrow \sqrt{2c+5} = 4 - c^2 \therefore c$ is solution of $\sqrt{2x+5} = 4 - x^2$

HW2-5

- HW: 5~10,15,19,26,30,42,45,54,55

102 5. 底下是加分題請只挑一題來作：

(a) (5 points) 求極限 $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = ?$

V (b) (5 points) 敘述中間值定理 (Intermediate Values Theorem)。

(c) (5 points) 寫出極限 $\lim_{x \rightarrow 2} f(x) = 5$ 的 $\epsilon - \delta$ 的定義。

103 3 (10 points) Show that the equation $\cos(x) + 1 = 2x$ has exactly one real solution.

part I: 證明有實根

$$\text{Let } f(x) = \cos(x) + 1 - 2x$$

$$f(0) = 2 - 0 = 2 > 0$$

$$f(2) = \cos 2 + 1 - 4 < 0$$

> 0 lies between $f(0)$ & $f(2)$

$\cos x$ is a trigonometric function

& $1 - 2x$ is a polynomial

$\therefore$ they are conti on $[0, 2]$

$\therefore f(x) = \cos x + 1 - 2x$ is also conti. on $[0, 2]$

$\therefore$ By I.V.T. $\exists c \in (0, 2)$ such that $f(c) = 0$.

$\Rightarrow \cos c + 1 - 2c = 0 \Rightarrow \cos c + 1 = 2c \therefore c$ is a solution of $\cos x + 1 = 2x$.

2-6 Limits Involving Infinity; Asymptotes of Graphs

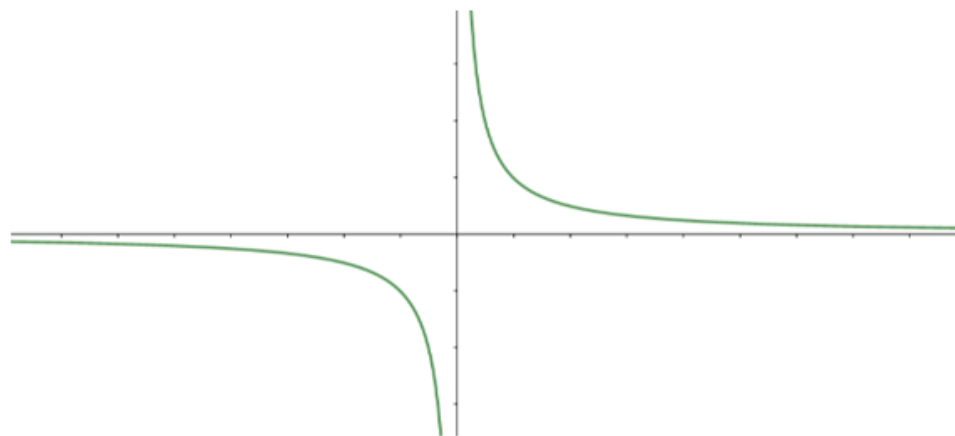
師大工教一

Observe $\lim_{x \rightarrow a} f(x) = L$ (i) $a = \pm \infty$ (ii) $L = \pm \infty$

※ **Finite Limits as $x \rightarrow \pm \infty$**

Ex1(p116): Show that (a) $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ (b) $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$

(0 與 $\pm \infty$ 互為倒數)



Theorem: Assume that c is a constant, $k > 0$, then $\lim_{x \rightarrow \pm\infty} \frac{c}{x^k} = 0$.

Ex3(p117): Find (a) $\lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2}$

$$\begin{aligned} (a) &= \lim_{x \rightarrow \infty} \frac{5 + \frac{8}{x} - \frac{3}{x^2}}{3 + \frac{2}{x^2}} \\ &= \frac{5}{3} \end{aligned}$$

(b) $\lim_{x \rightarrow -\infty} \frac{11x + 2}{2x^3 - 1}$

$$\begin{aligned} (b) &= \lim_{x \rightarrow -\infty} \frac{\frac{11}{x^2} + \frac{2}{x^3}}{\frac{2}{x^3} - \frac{1}{x^3}} \\ &= 0 \end{aligned}$$

※Horizontal Asymptotes

Definition: A line $y = b$ is a **horizontal asymptote** of the graph of a function

$y = f(x)$ if either $\lim_{x \rightarrow \infty} f(x) = b$ or $\lim_{x \rightarrow -\infty} f(x) = b$.

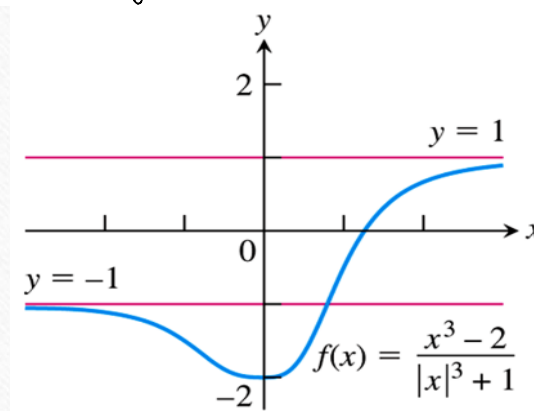
$$y = b \quad \lim_{x \rightarrow \pm\infty} f(x) = b$$

Ex4(p118) Find the horizontal asymptotes of the graph of $f(x) = \frac{x^3 - 2}{|x|^3 + 1}$.

$$\lim_{x \rightarrow \infty} \frac{x^3 - 2}{|x|^3 + 1} = \lim_{x \rightarrow \infty} \frac{1 - \frac{2}{x^3}}{1 + \frac{1}{x^3}} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{x^3 - 2}{|x|^3 + 1} = \lim_{x \rightarrow -\infty} \frac{1 - \frac{2}{x^3}}{-1 + \frac{1}{x^3}} = -1$$

$\therefore y = 1$ and $y = -1$ are horizontal asymptotes.



※ Oblique Asymptotes (slant line asymptotes)

斜漸近線 (用在有理函數, 分子次數 = 分母次數)

If the degree of the numerator of a rational function is 1 greater than the degree of the denominator, the graph has an **oblique** or **slant line asymptote**.

Ex10(p120) Find the oblique asymptote of the graph of $f(x) = \frac{x^2 - 3}{2x - 4}$.

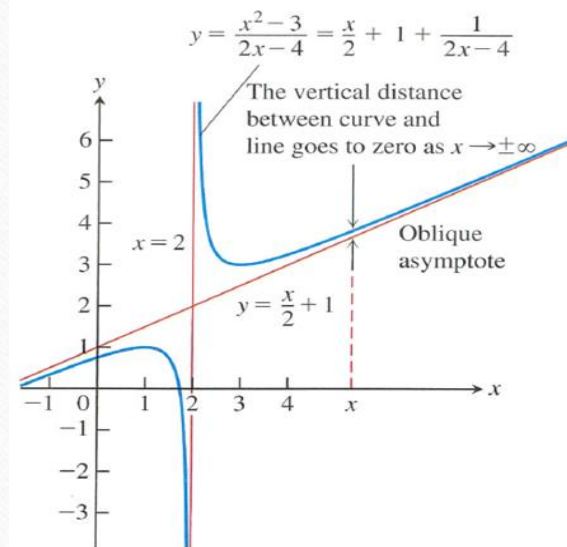
$$\begin{array}{r} - 4 \overline{) 1 + 0 - 3} \\ \underline{1 - 2} \\ 2 - 3 \\ \underline{2 - 4} \\ - 4 \end{array}$$

$$\frac{x^2-3}{2x-4} = \frac{1}{2}x+1 + \frac{1}{2x-4}$$

$$\lim_{x \rightarrow \infty} f(x) - \left(\frac{1}{2}x + 1\right) = \lim_{x \rightarrow \infty} \frac{1}{2x-4} = 0$$

$$\text{III} \quad \lim_{x \rightarrow \infty} f(x) - \left(\frac{1}{2}x + 1\right) = 0$$

$\therefore y = \frac{1}{5}x + 1$ is a slant line asym.



※Infinite Limits

Ex: Find the following limits:

$$(i) \lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$$

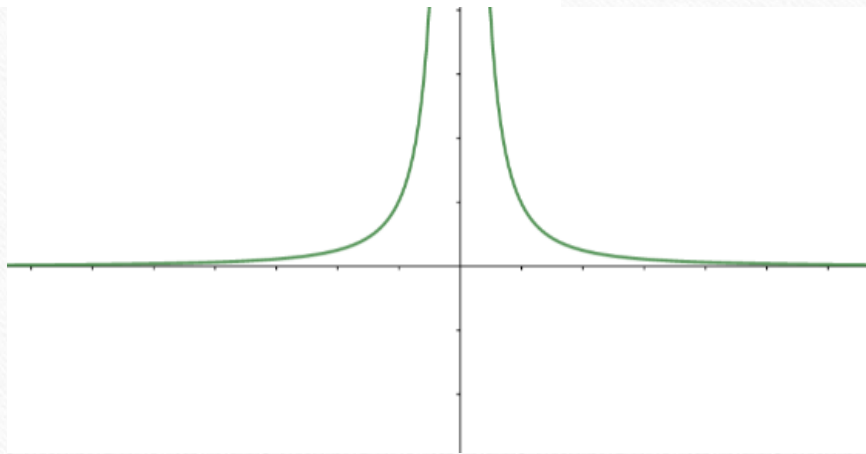
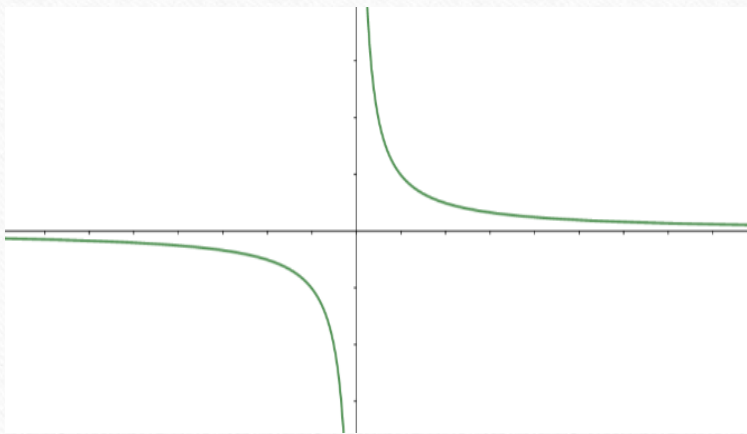
$$(ii) \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

$$(iii) \lim_{x \rightarrow 0} \frac{1}{x} \text{ DNE.}$$

$$(iv) \lim_{x \rightarrow 0^+} \frac{1}{x^2} = \infty$$

$$(v) \lim_{x \rightarrow 0^-} \frac{1}{x^2} = \infty$$

$$(vi) \lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$



※Vertical Asymptotes (分母 = 0 就有鉛直Asym.)

鉛直 漸近線

A line $x = a$ is a **vertical asymptote** of the graph of the function $y = f(x)$ if

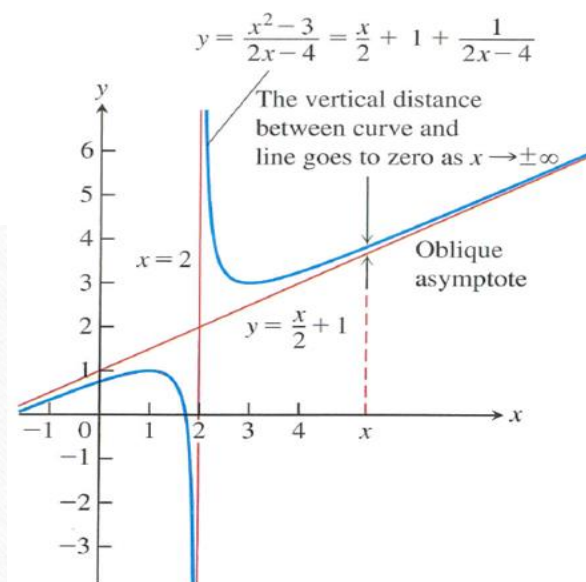
either $\lim_{x \rightarrow a^+} f(x) = \pm\infty$ or $\lim_{x \rightarrow a^-} f(x) = \pm\infty$.

※Dominant Terms

Ex10: $f(x) = \frac{x^2 - 3}{2x - 4} = \left(\frac{x}{2} + 1\right) + \left(\frac{1}{2x - 4}\right)$

We say that $\frac{x}{2} + 1$ dominates when x approach ∞ or $-\infty$.
控制

We say that $\frac{1}{2x - 4}$ dominates when x approach 2.



HW2-6

- HW: 13,14,28,42,53,65,67,80,105,108

111

2. (18 pts) Find the limit if it exists.

$$(a) \lim_{x \rightarrow 0} \frac{\sin 3x - 3x + x^2}{\sin x \sin 2x}$$

$$\text{V} (b) \lim_{x \rightarrow \infty} \frac{3x}{5x + 2 \sin x}$$

$$(c) \lim_{x \rightarrow 0} \left(\frac{1}{x\sqrt{1+x}} - \frac{1}{x} \right)$$

$$(b) \lim_{x \rightarrow \infty} \frac{\frac{3x}{x}}{\frac{5x}{x} + \frac{2 \sin x}{x}} = \lim_{x \rightarrow \infty} \frac{3}{5 + 2 \cdot \frac{\sin x}{x}} = \frac{3}{5}$$

$$\begin{array}{l} -1 \leq \sin x \leq 1 \\ \frac{-1}{|x|} \leq \frac{\sin x}{x} \leq \frac{1}{|x|} \\ \lim_{x \rightarrow \infty} \frac{-1}{|x|} = 0 = \lim_{x \rightarrow \infty} \frac{1}{|x|} \end{array} \quad \begin{array}{l} \text{By sandwich theorem} \\ \lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0 \end{array}$$

$$103 \quad 4 \quad \text{Let } f(x) = \frac{2x^3}{x^2-1}$$

(有斜 asym. 就不會有水平 asym.)

- (a) (10 points) find horizontal asymptotes (水平漸近線), vertical asymptotes (垂直漸近線) and slant line asymptotes (斜漸近線) of $f(x)$,
- (b) (10 points) find all critical points (臨界點) of $f(x)$.

$$(a) \frac{2x^3}{x^2-1} = 2x + \frac{2x}{x^2-1}$$

$$\lim_{x \rightarrow \infty} \left(\frac{2x^3}{x^2-1} - 2x \right) = \lim_{x \rightarrow \infty} \frac{2x}{x^2-1} = 0$$

$\therefore y=2x$ is oblique asymptote.

$$x^2-1 = (x+1)(x-1) \quad \lim_{x \rightarrow 1} \frac{2x^3}{x^2-1} = \infty \left(\frac{+}{0} \right)$$

$$\lim_{x \rightarrow -1} \frac{2x^3}{x^2-1} = \infty \left(\frac{-}{0} \right)$$

$\Rightarrow x=1, x=-1$ are vertical asymptotes.

105分 6. Let

$$f(x) = \ln \frac{3(x^2 - x - 2)}{x^2 - 4}.$$

- (a) (6 points) Find the domain of the function $f(x)$. $(-\infty, -2) \cup (-1, 2) \cup (2, \infty)$
(b) (12 points) Find the vertical and horizontal asymptotes for the graph of the function $f(x)$.

$y = \ln 3$ horizontal asym.

$x = -2, x = -1$, vertical asymptotes.